UNIVERSITY OF CAMBRIDGE

Bridging the Gap Between Supernova Cosmology and Astrophysics

Matt Grayling

ioa

Kavli Institute for Cosmology Cambridge KICC

erc European Research Council Established by the European Commission

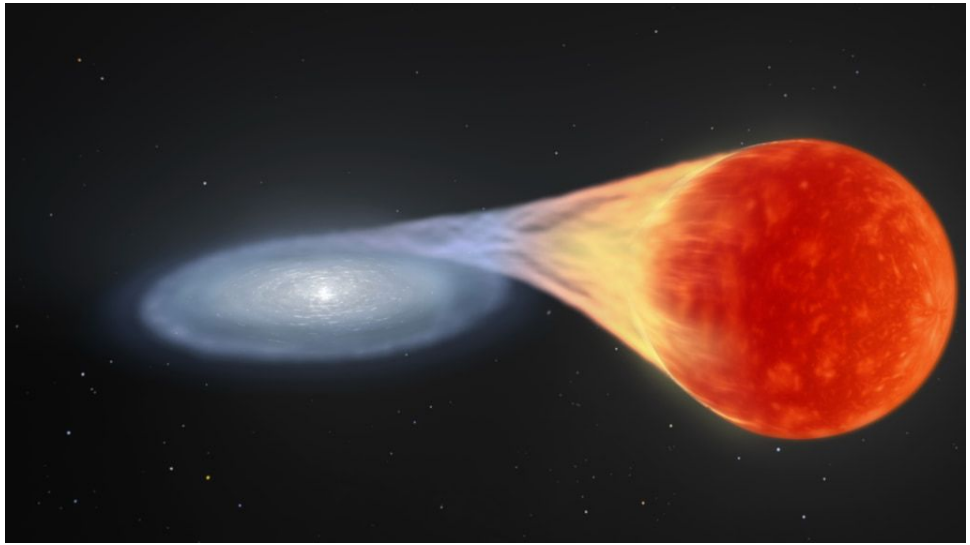
Talk Overview

1) Introduction to supernovae
2) BayeSN: Bringing more astrophysics to supernova cosmology
3) BayeSN-TD: A brief look at gravitationally-lensed supernovae

Part 1: Why Care About Supernovae?

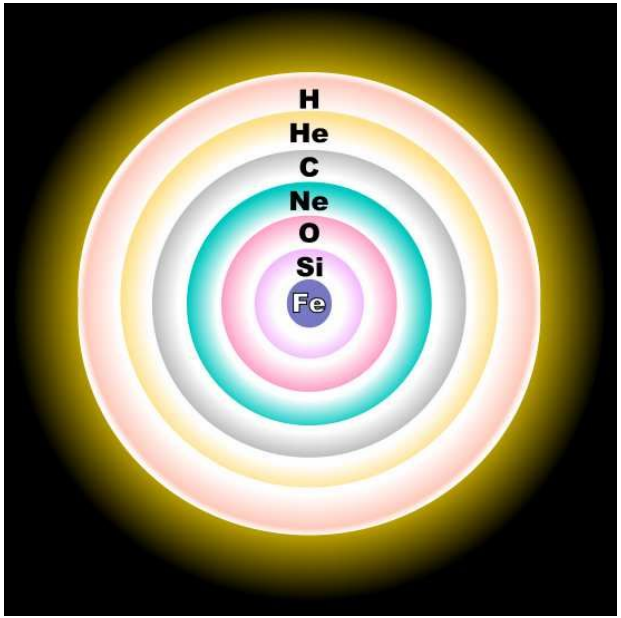
Supernovae

Type Ia



Credit: Adriana Manrique Gutierrez svs.gsfc.nasa.gov

Core-collapse



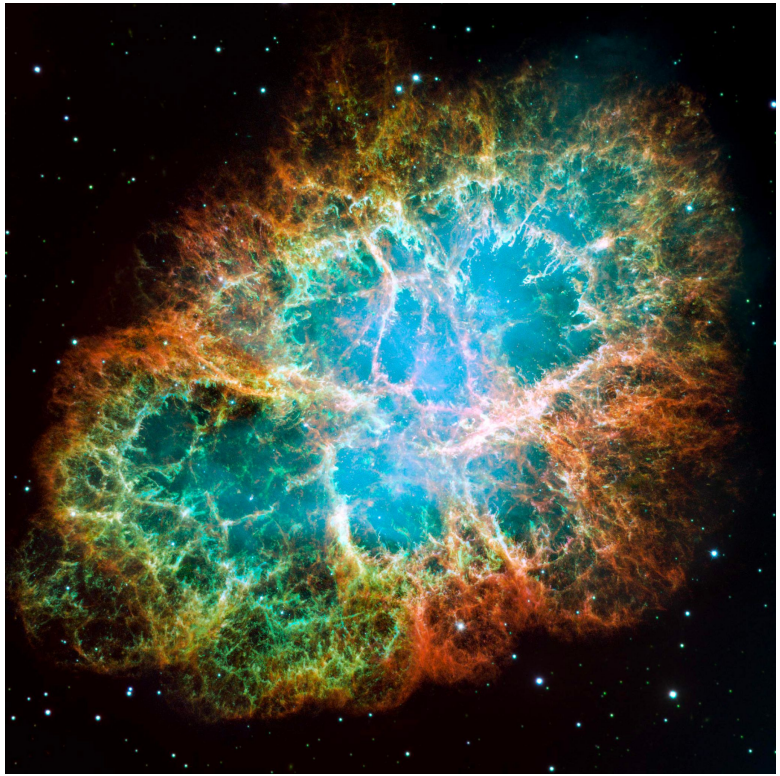
Why are supernovae interesting?

Astrophysics

- Create and distribute heavy elements
- Release energy into their surroundings
- Environment to study extreme explosion physics

Cosmology

- Type Ia supernovae can be used to measure distances



Credits: NASA, ESA, J. Hester and A. Loll (Arizona State University)

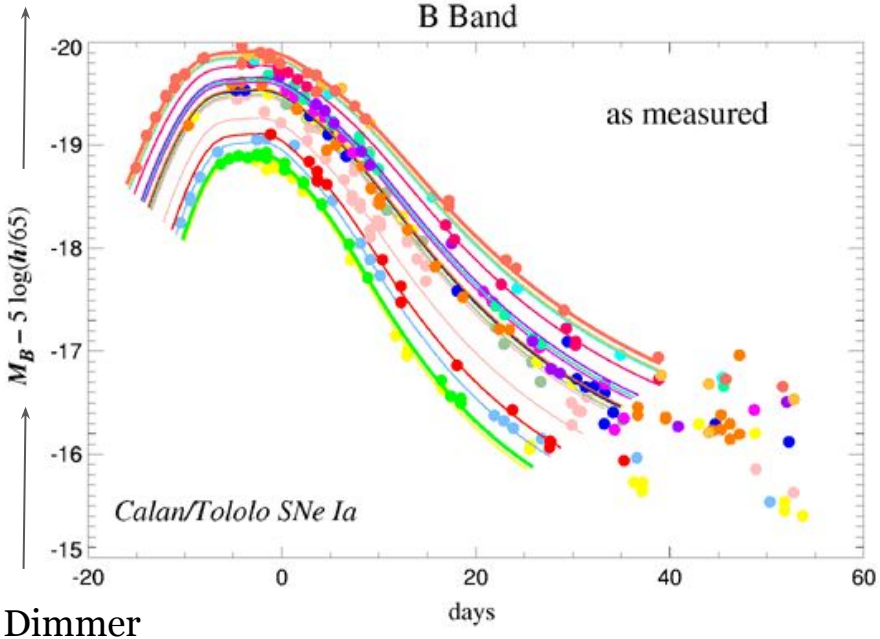
Standard Candles

$$F = \frac{L}{4\pi d^2}$$

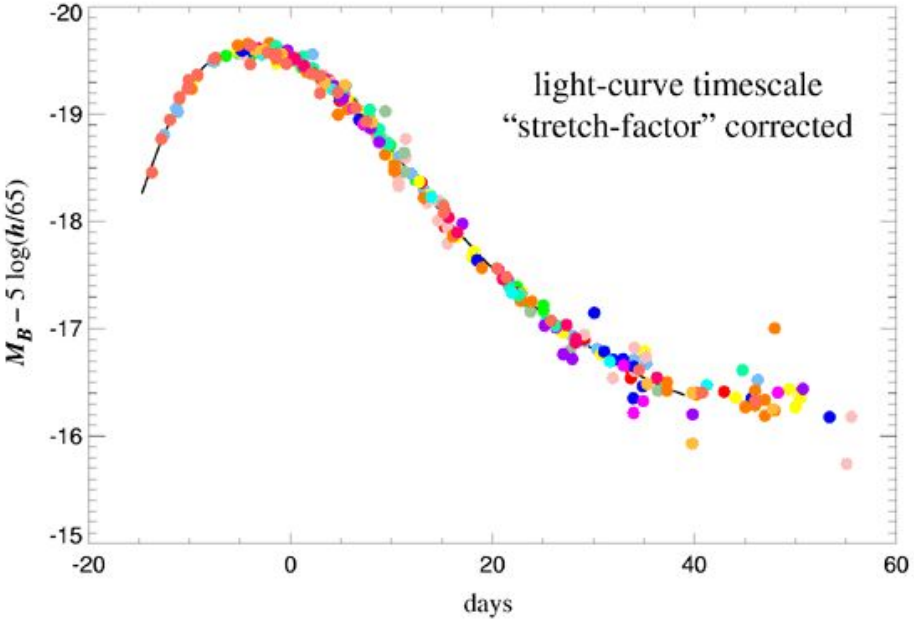
$$\mu = m - M$$

Standard(isable) Candles

Brighter



Dimmer

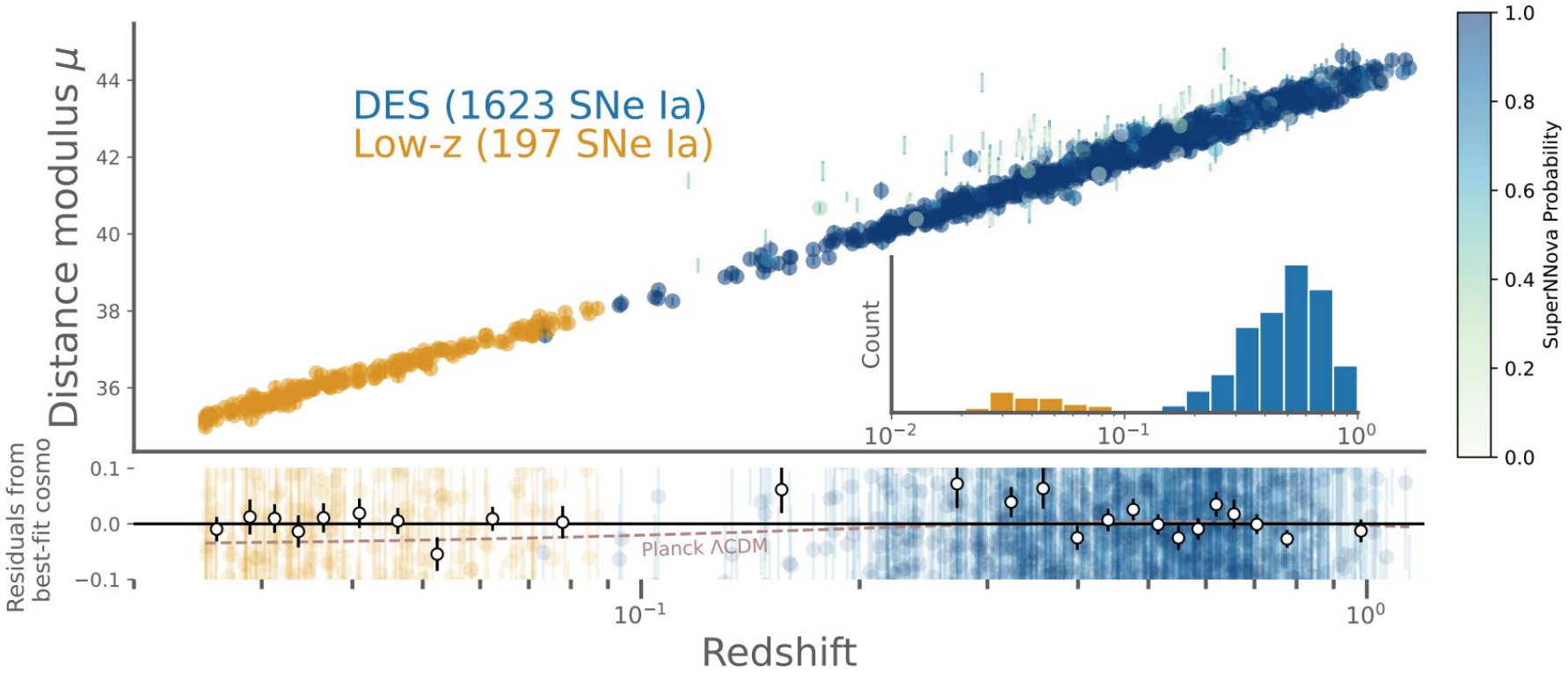


From Supernova Cosmology Project

Calculating Distance

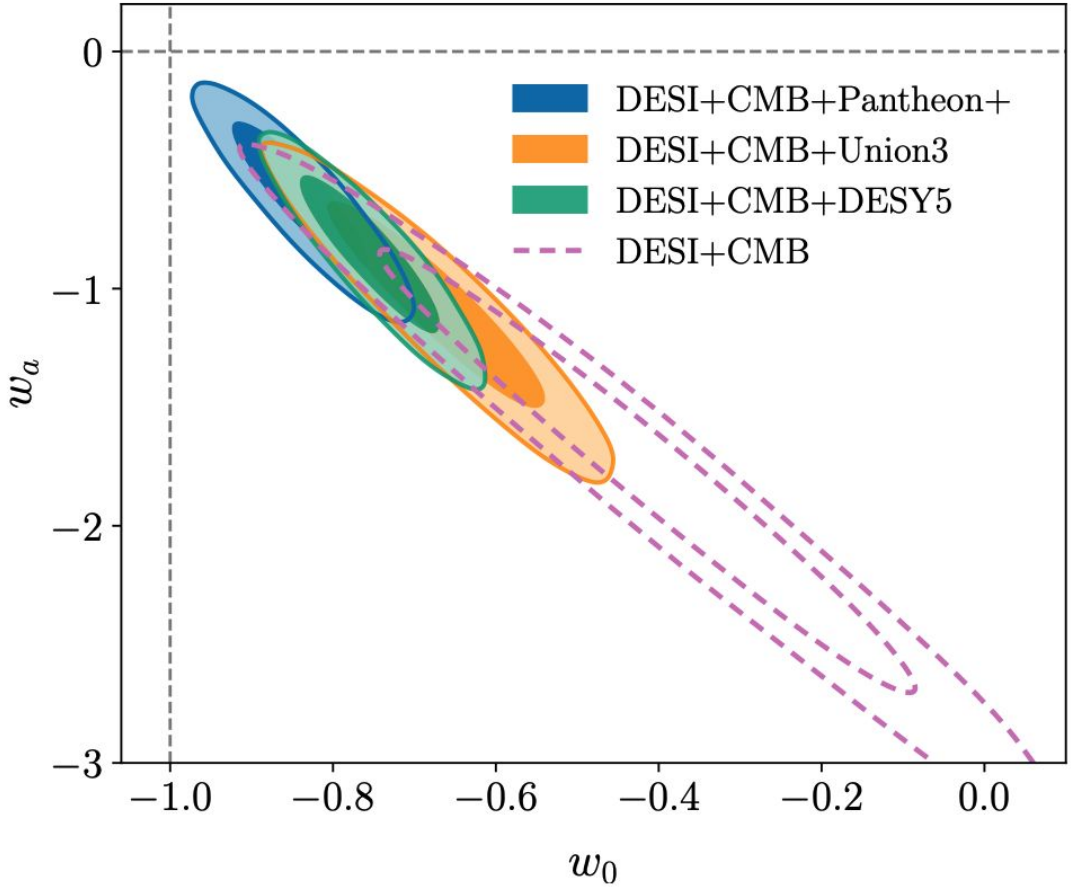
$$\mu^s = m_B^s - M + \alpha x_1^s - \beta c^s$$

The Hubble Diagram



Popovic+2025

Crisis in Cosmology?



DESI Collaboration 2025

Supernova cosmology is easy, right?

The Impact of Dust

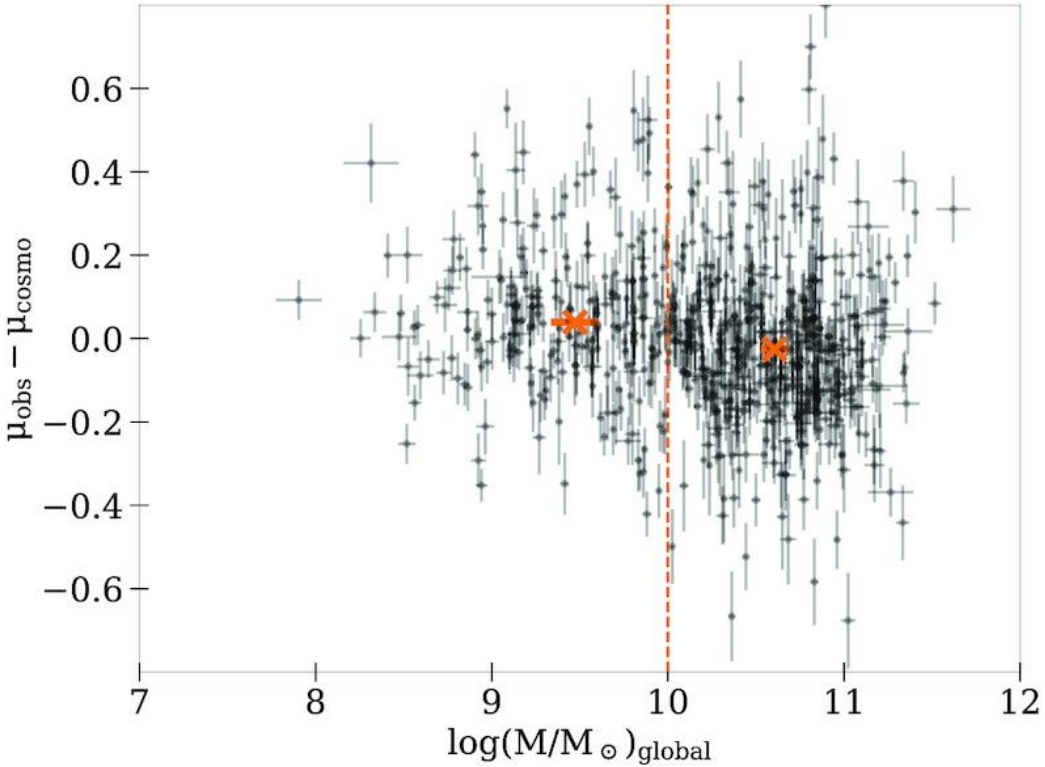
$$\mu^s = m_B^s - M + \alpha x_1^s - \beta c^s$$

$$c^s = c_{\rm int}^s + E(B-V)^s$$

- Colour - one parameter for two separate effects
- Correctly handling dust is key to uncover intrinsic SN properties and accurately estimate distances

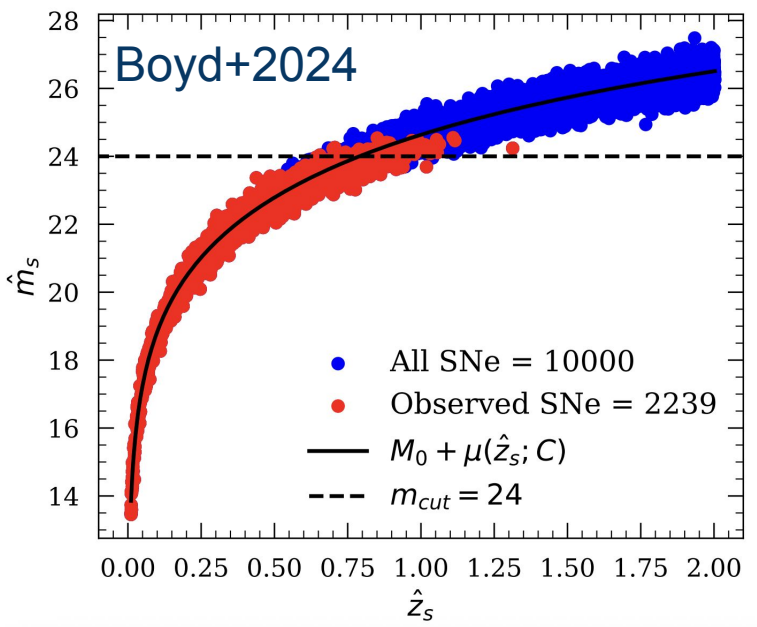
The Mass Step

- SN Ia properties depend on environment
- Just different dust properties?
  - SN populations identical, just appear different?
- Or intrinsic differences?
  - Interesting physical implications, different progenitor channels?

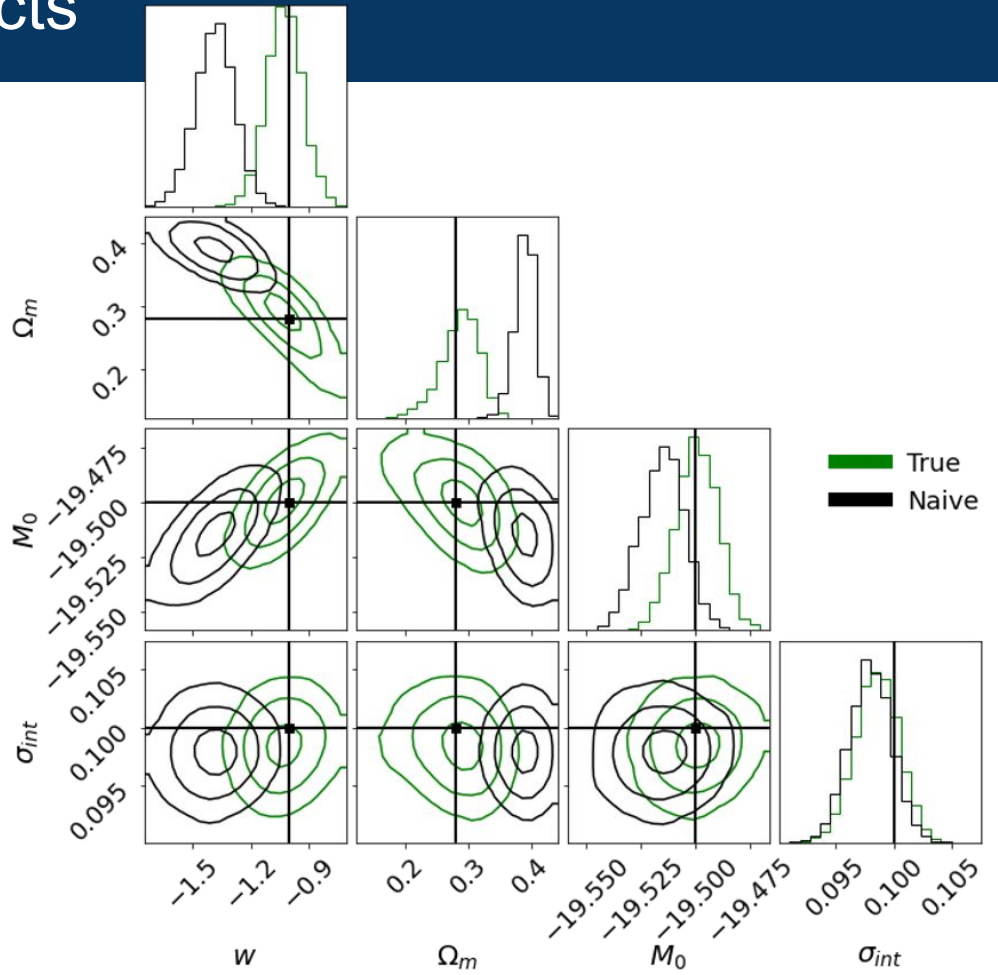


Kelsey+2023

The Impact of Selection Effects



- Selection effects characterised by simulations
- Simulations require a model; unknown astrophysics is the largest systematic uncertainty in SN cosmology



Supernova cosmology is inseparably linked to supernova astrophysics

Part 2: BayeSN

Recipe for a Type Ia Supernova Model

Type Ia SN SED

=

+

+

+



Recipe for a Type Ia Supernova Model

Type Ia SN SED

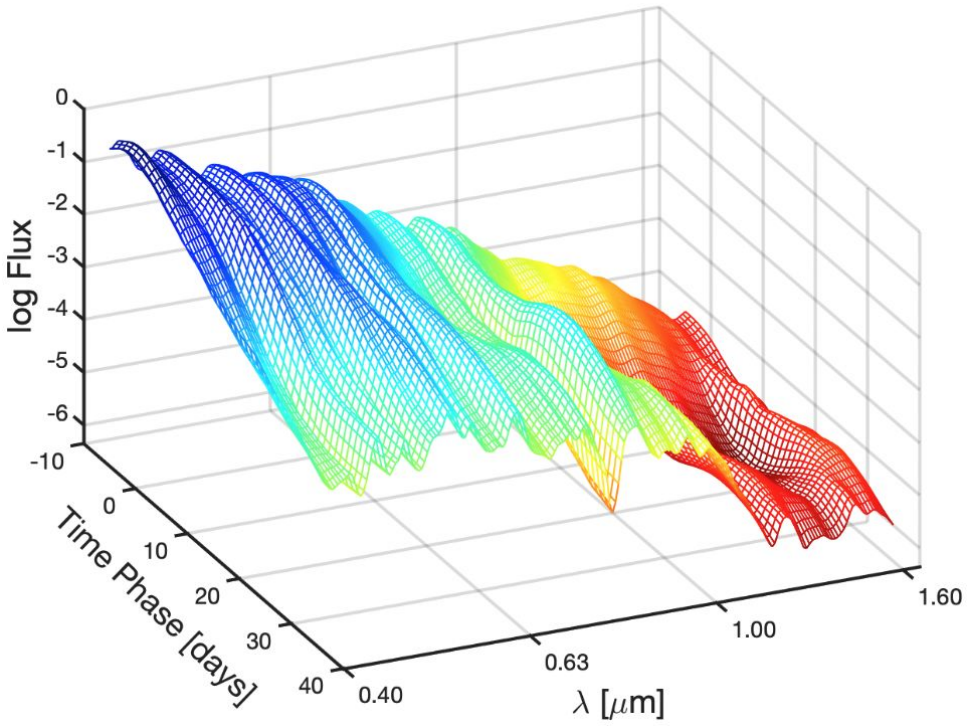
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Mean Intrinsic SED

+

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Recipe for a Type Ia Supernova Model

Type Ia SN SED

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Mean Intrinsic SED

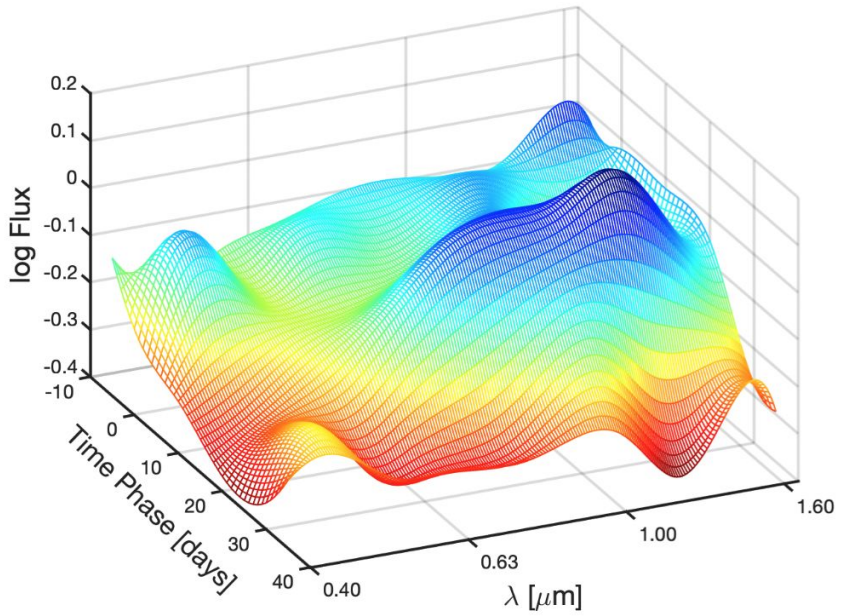
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Effect of Stretch

+

+

$\theta_1 \times$



Recipe for a Type Ia Supernova Model

Type Ia SN SED

=

Mean Intrinsic SED

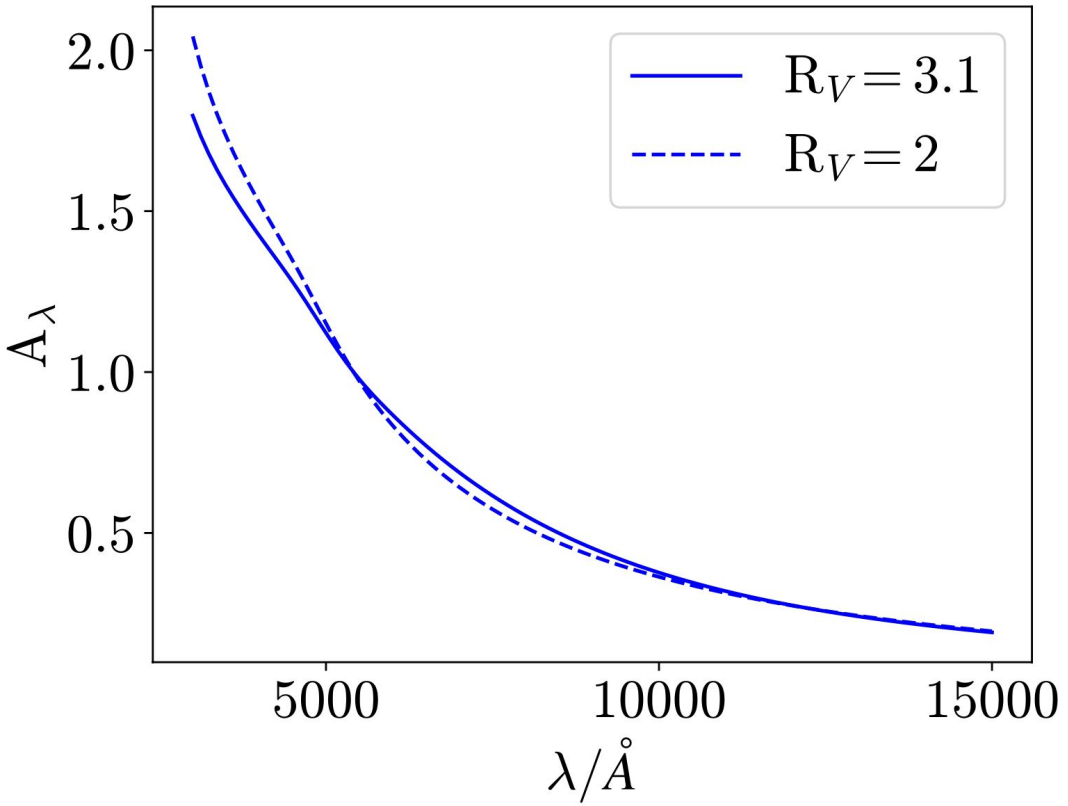
+

Effect of Stretch

+

Dust Extinction

+



Recipe for a Type Ia Supernova Model

Type Ia SN SED

=

Mean Intrinsic SED

+

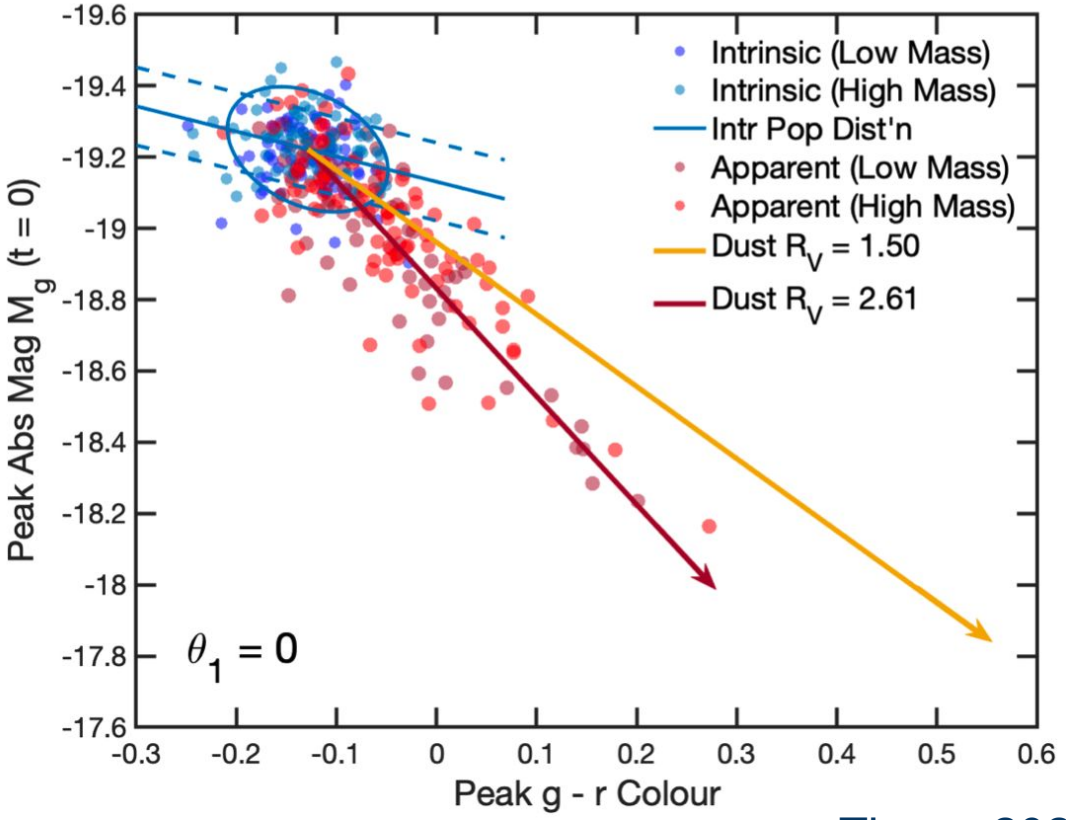
Effect of Stretch

+

Dust Extinction

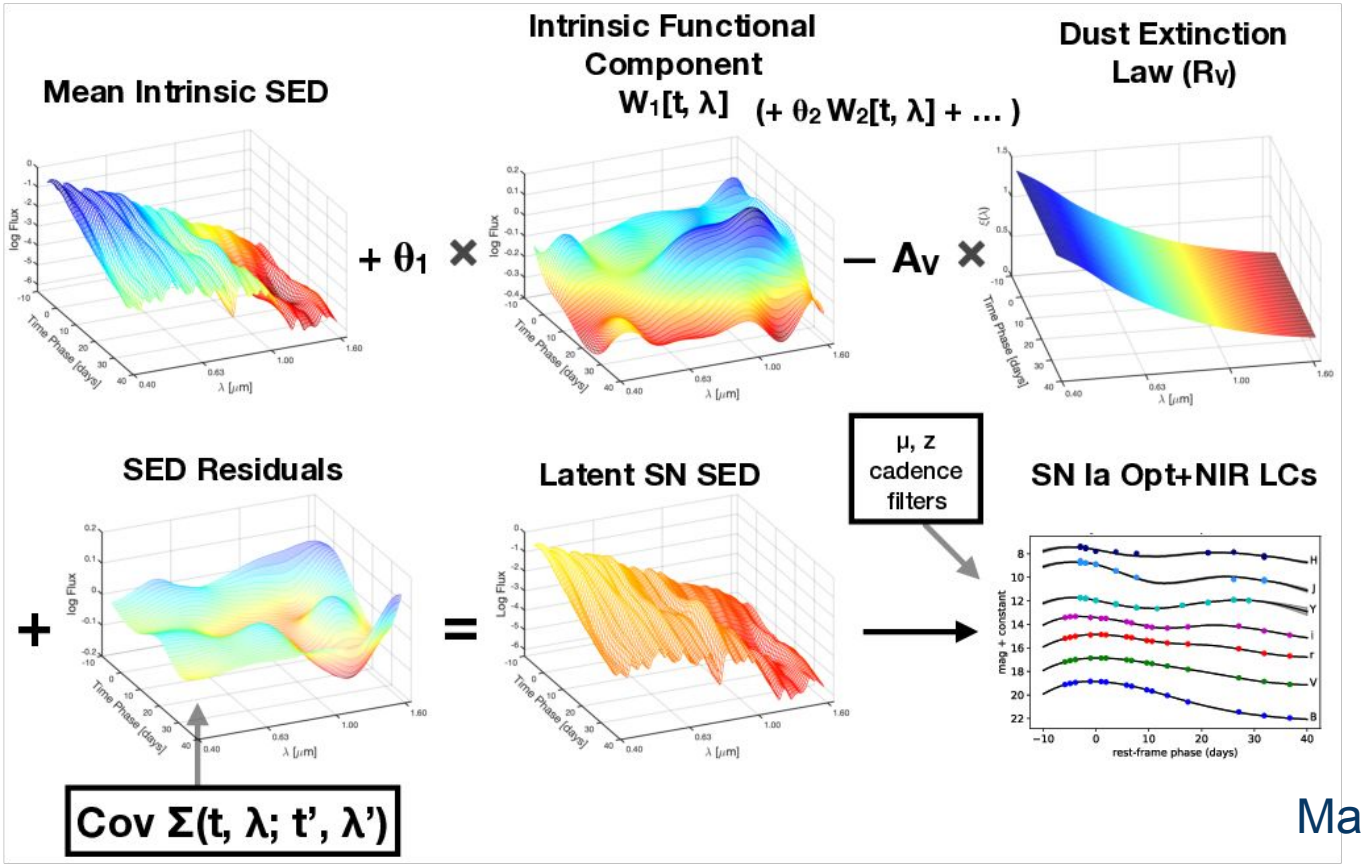
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Intrinsic Colour



Thorp+2021

The BayeSN Model



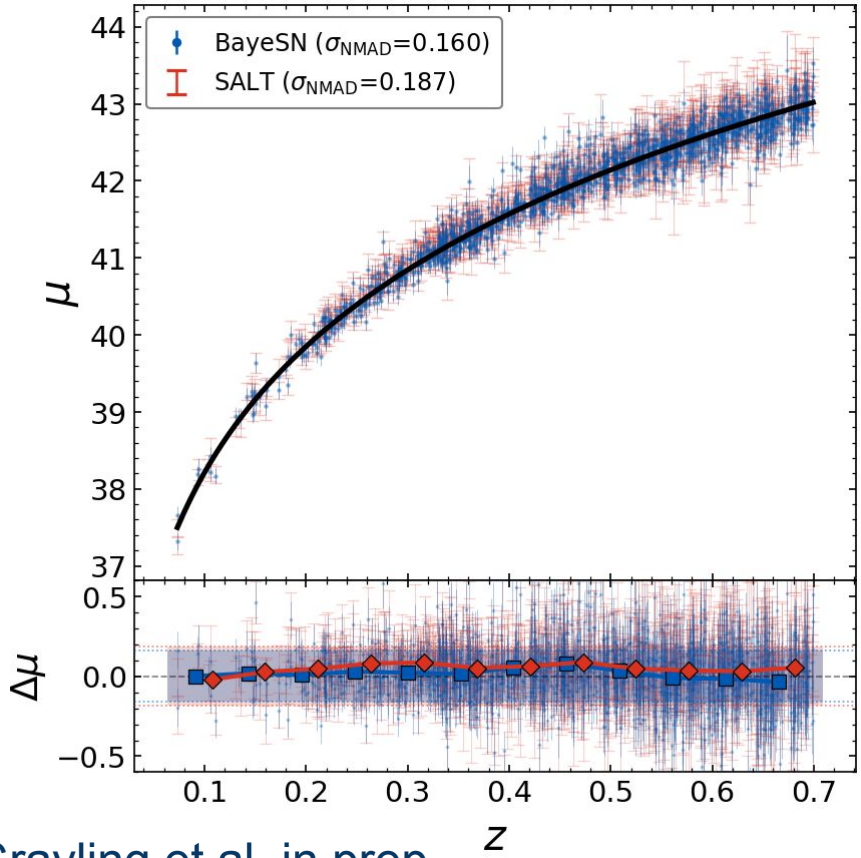
Mandel+2020

The BayeSN Model

- BayeSN can be used to infer distances to type Ia supernovae, and also properties of the population e.g. dust
- Using GPUs, you can do Bayesian inference with expensive likelihoods/thousands of parameters very fast!

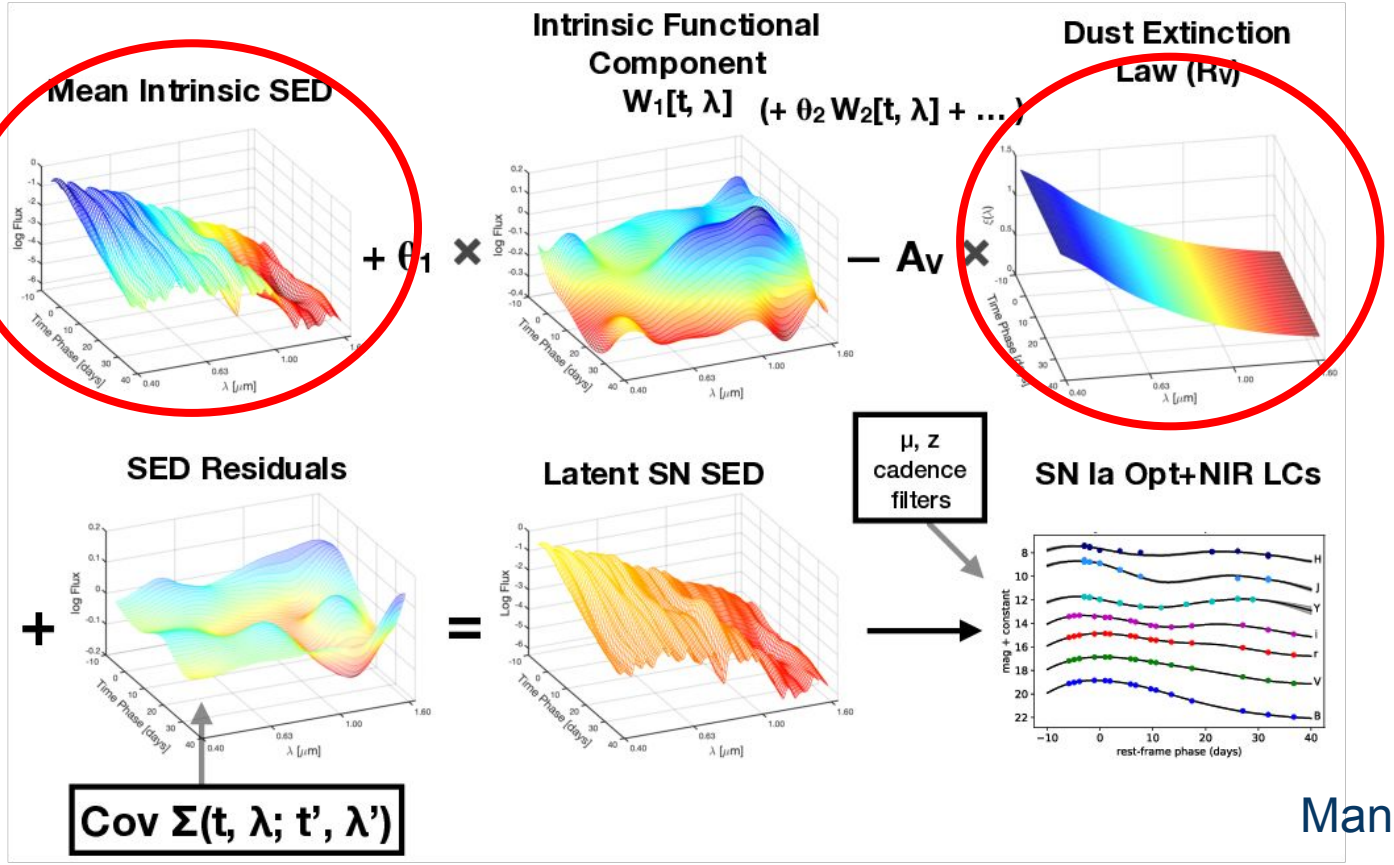






Grayling et al. in prep

Investigating the Mass Step



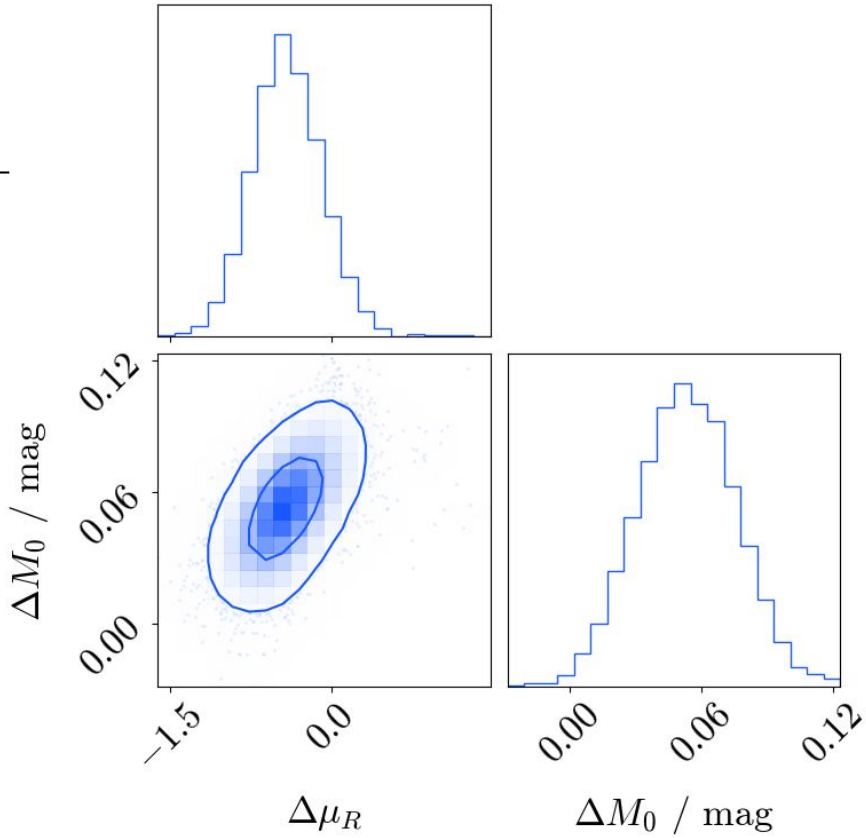
Mandel+2020

Cause of the Mass Step

Sample	Intrinsic Mass Step / mag
Combined*	0.049 ± 0.019
DES5YR	0.053 ± 0.022

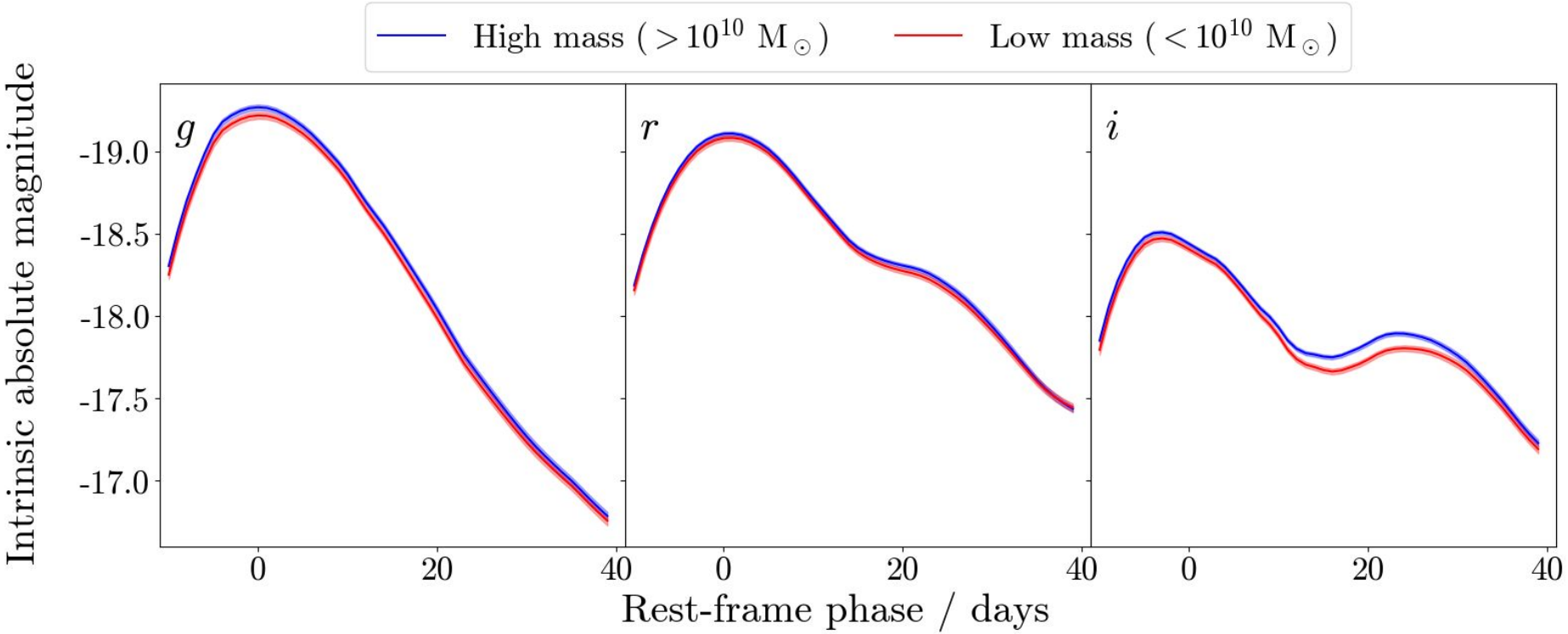
*Combined = DES3YR+Foundation+PS1MD

Dust may play a role, but BayeSN supports an intrinsic contribution to the mass step



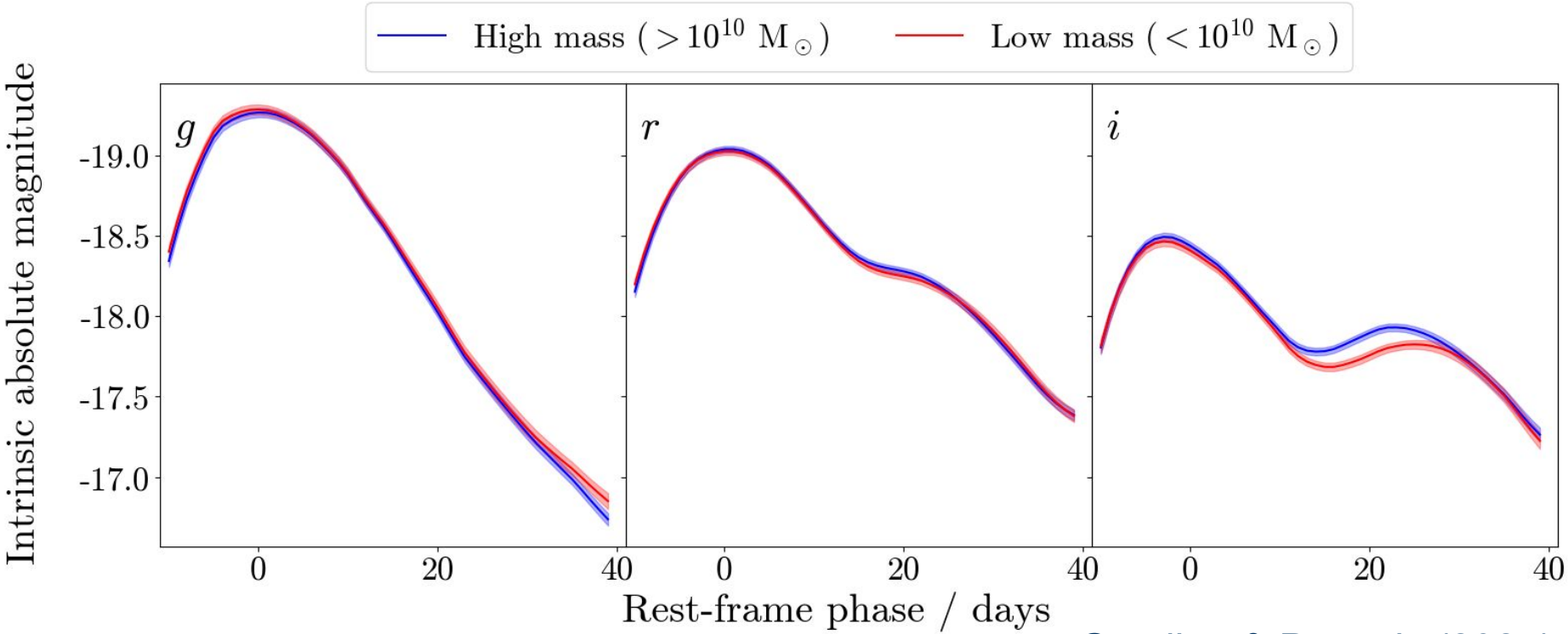
Grayling & Popovic (2025)

DES3YR + Foundation + PS1MD



Grayling+2024

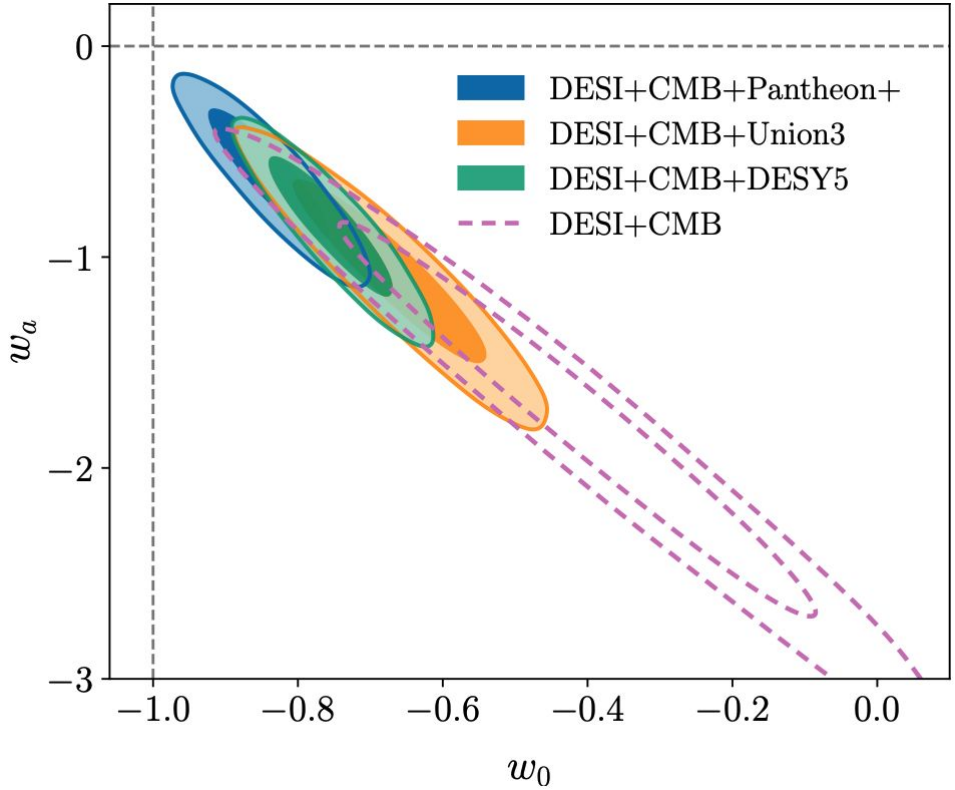
DES5YR



Grayling & Popovic (2024)

What Does This Tell Us?

- *Something* is causing an intrinsic difference between SNe Ia in different environments
- This seems to be related to the i-band secondary maximum, need to investigate this further
- Maybe links to progenitor age
- Significant for astrophysics, also very significant for cosmology



DESI Collaboration 2025

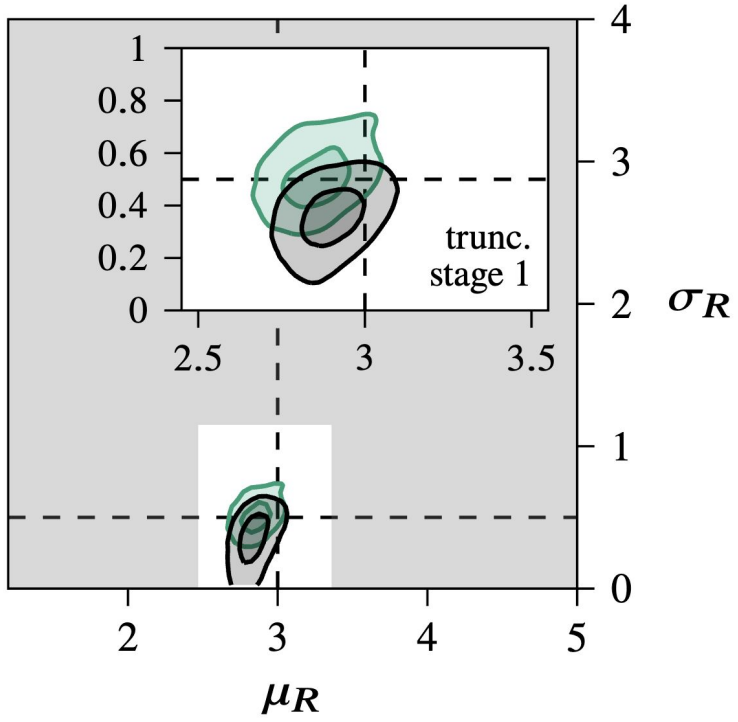
Plans for the Future

Bridging the gap between supernova cosmology and astrophysics:

- Expand from BayeSN, work towards constraining everything at once
- Joint inference of cosmology, dust, mass step etc., robust cosmological constraints marginalised over unknown astrophysics

Simulation-based inference (SBI) is the key:

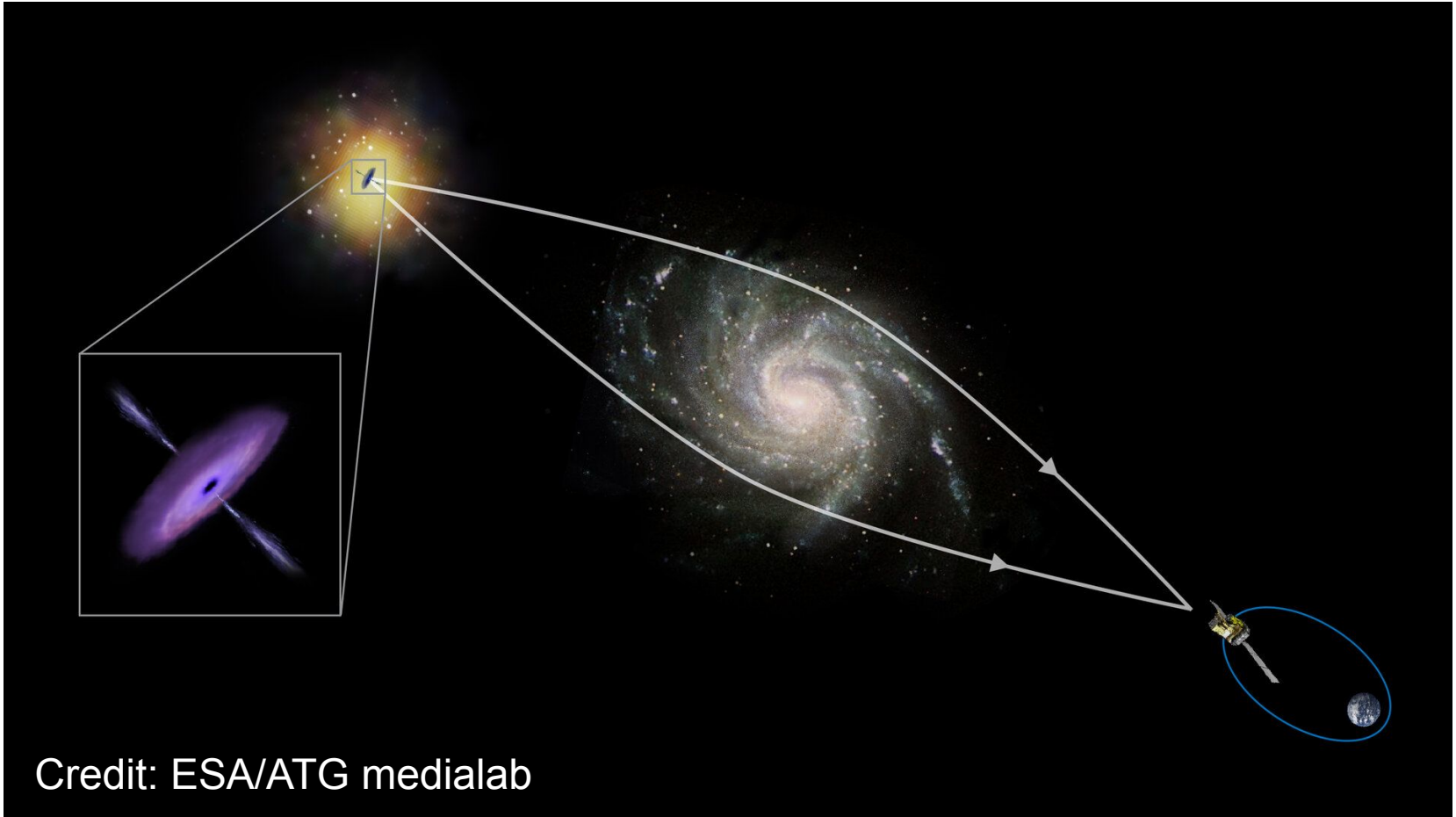
- Effectively, make lots of simulations and see which ones look like your data
- Neural networks learn mapping between parameters and data, making this fast and scalable



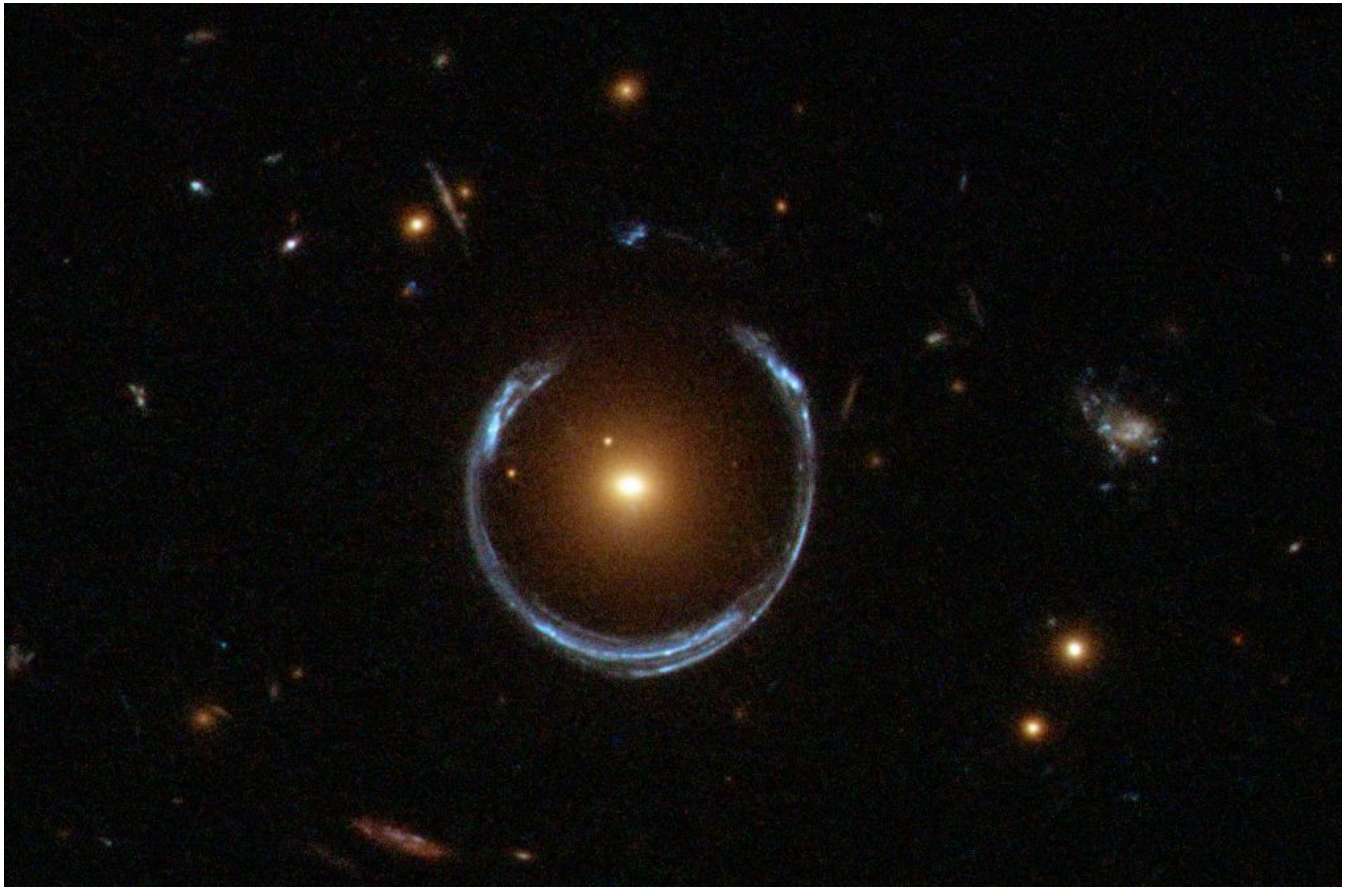
Karchev, Grayling et al. (2024)

Part 3: BayeSN-TD for lensed supernovae

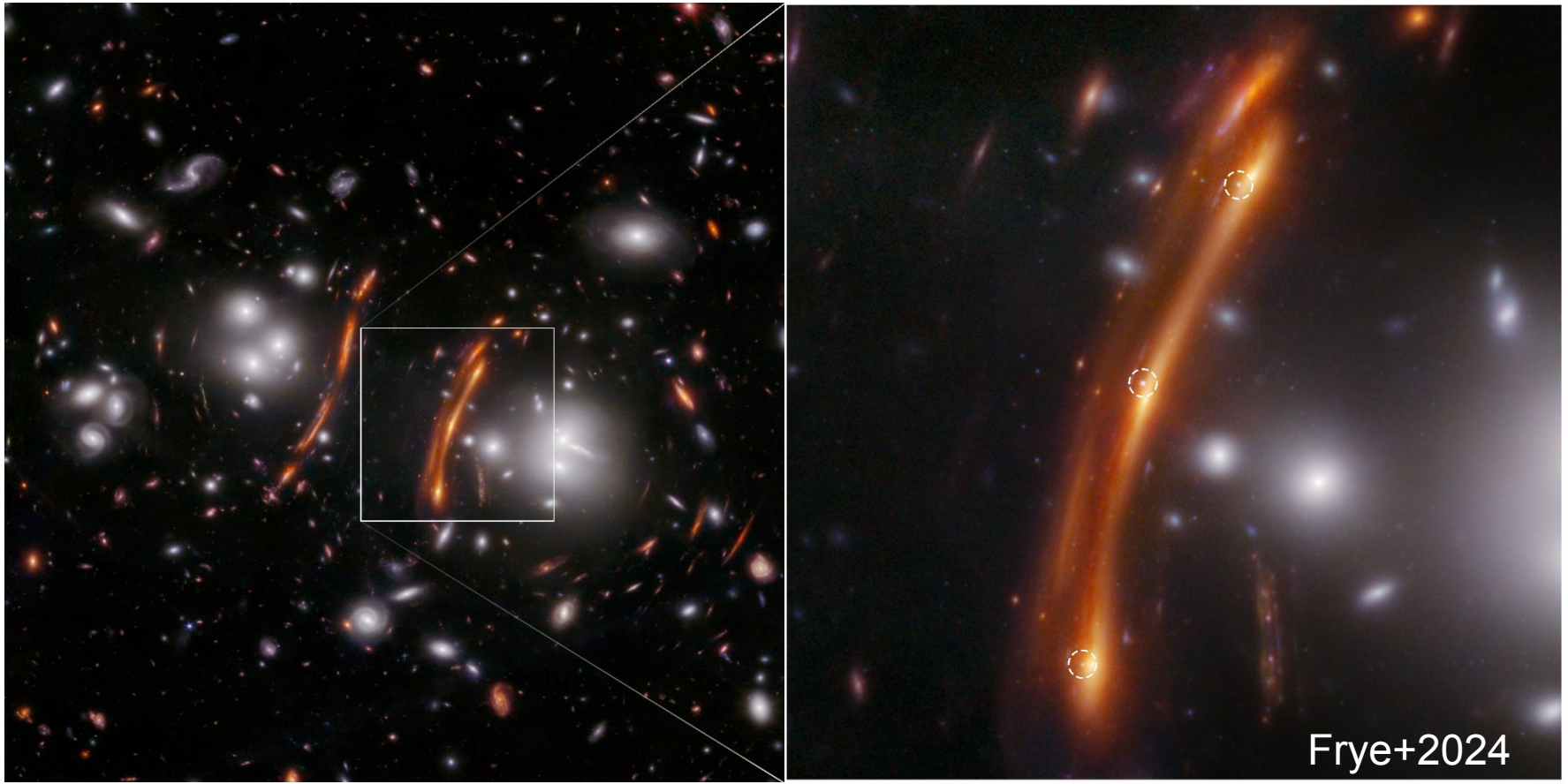
Gravitational Lensing



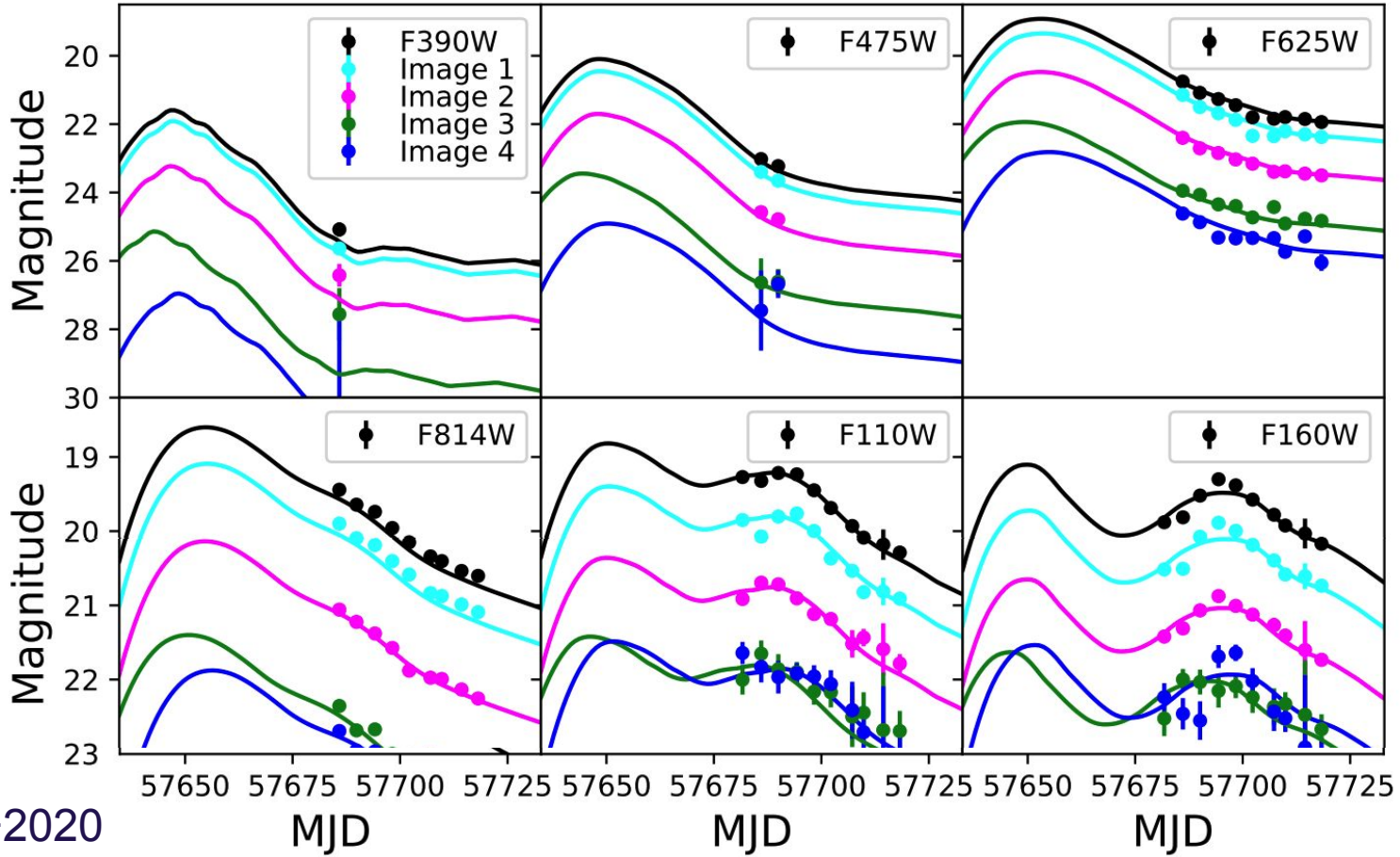
Gravitational Lensing



Gravitational Lensing

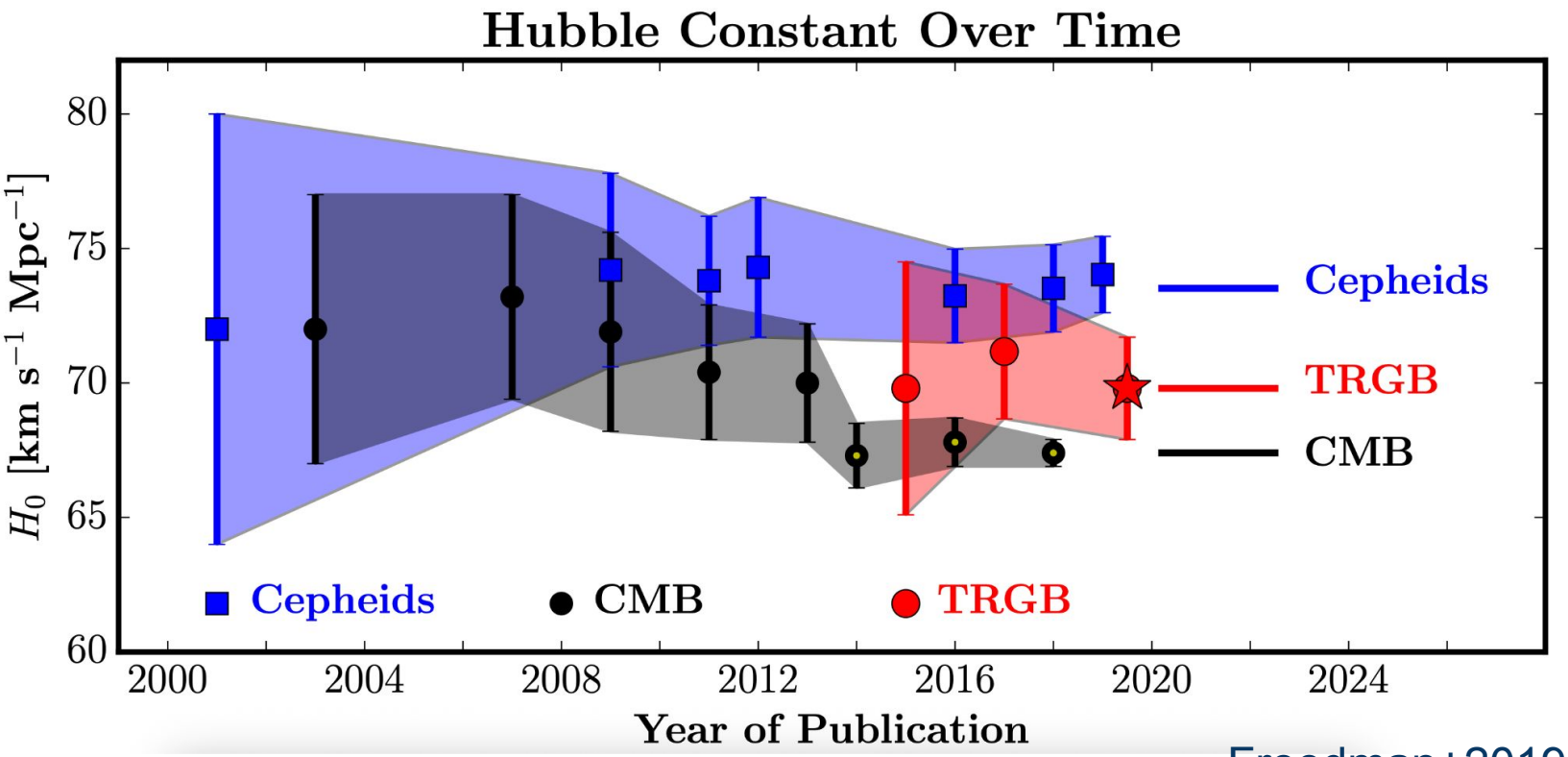


Lensed Supernova Light Curves



Dhawan+2020

The Hubble Tension



Freedman+2019

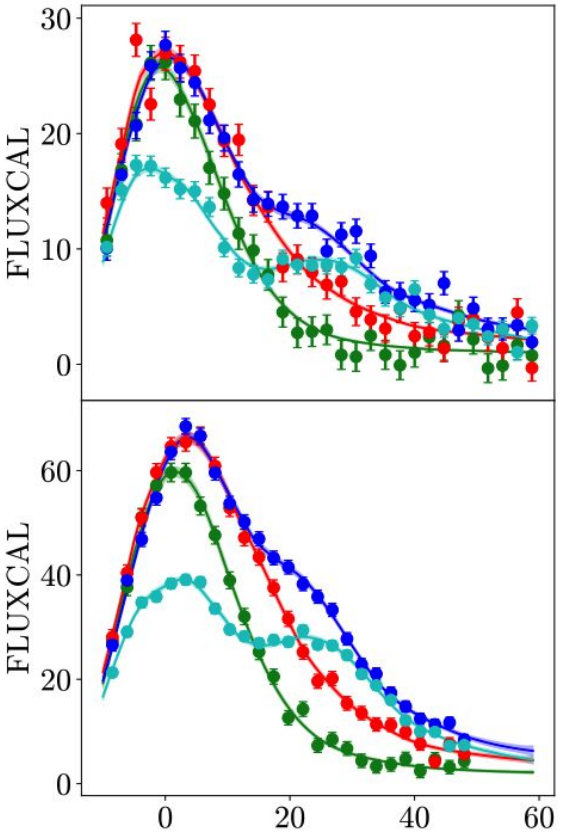
Inferring H0 with Lensed Supernovae

Time delay + Lens model = H_0

*Warning: May be simplified

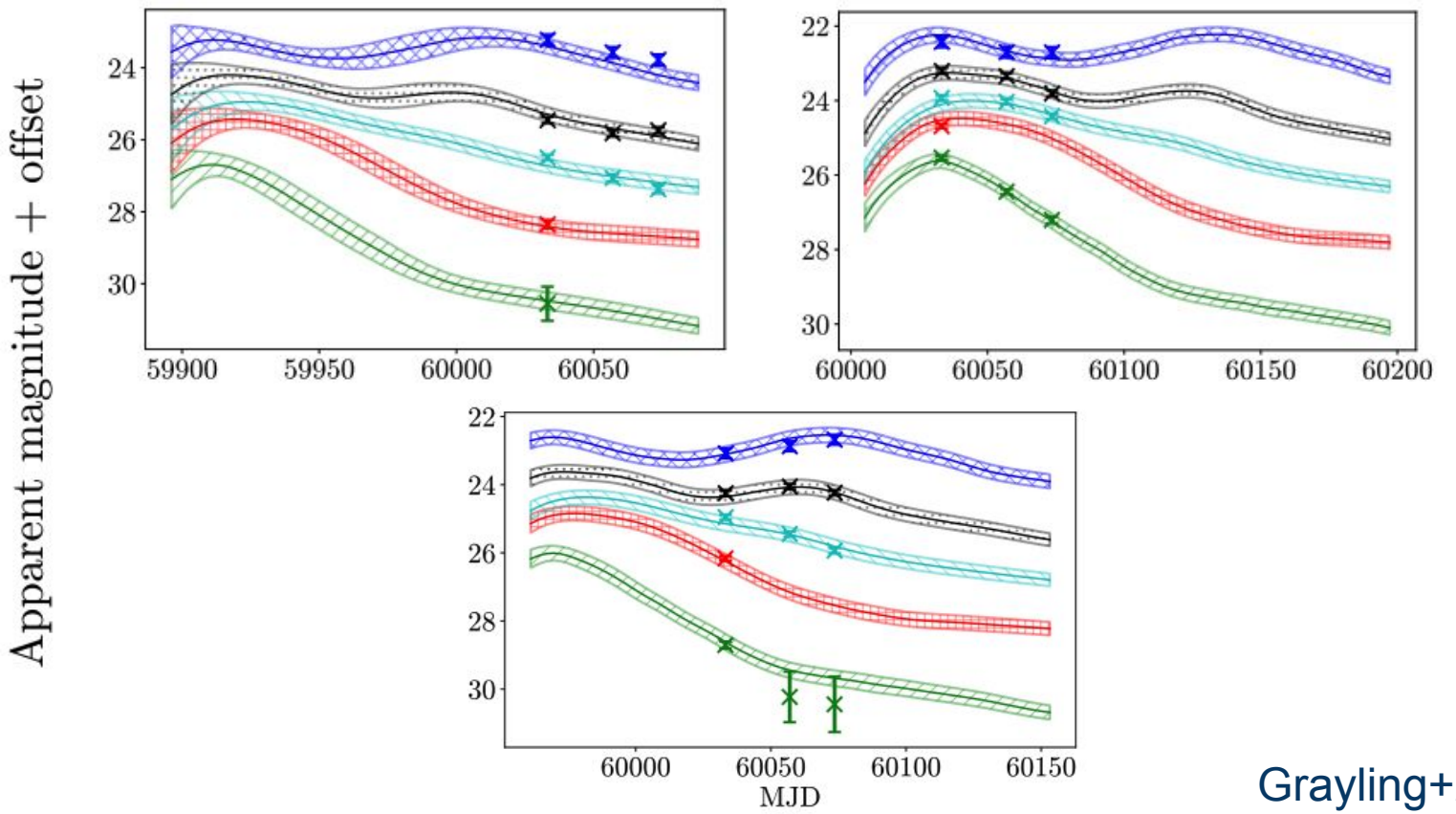
BayeSN-TD

- Modify BayeSN for lensed SNe Ia
- Some parameters shared between images, some unique for each image
  - SN parameters shared e.g. stretch
  - Per-image peak MJD, distance
- BayeSN-TD can be used to infer time delays and magnifications for each image
- We have validated BayeSN-TD on simulations, finding it gives robust estimates and uncertainties



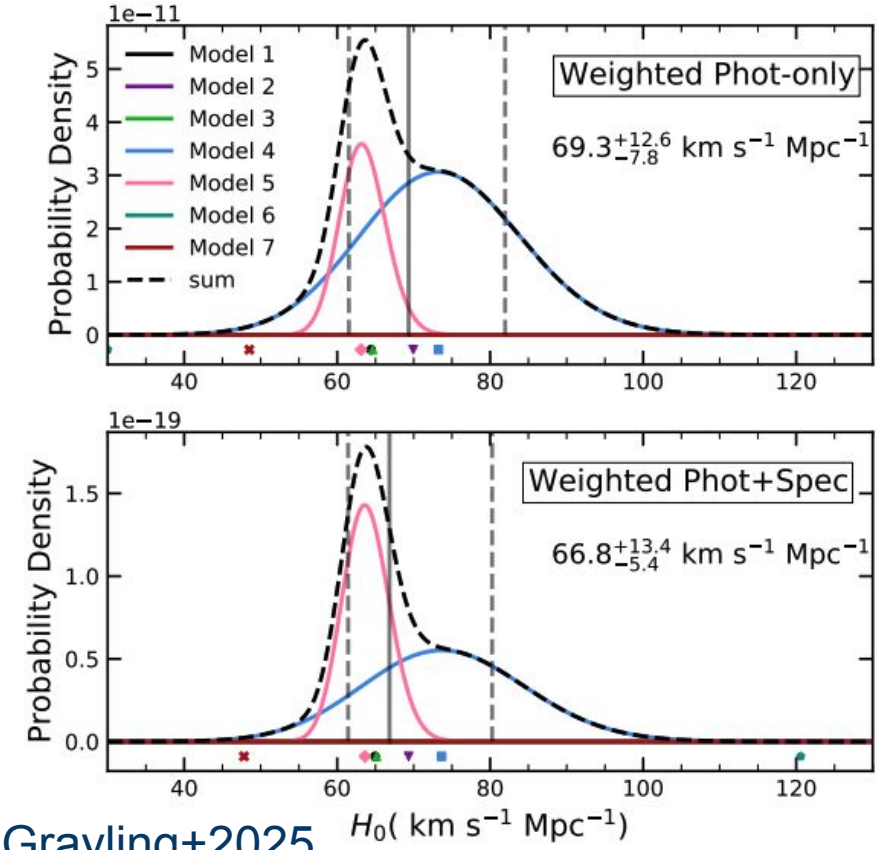
Grayling+2025

SN H0pe

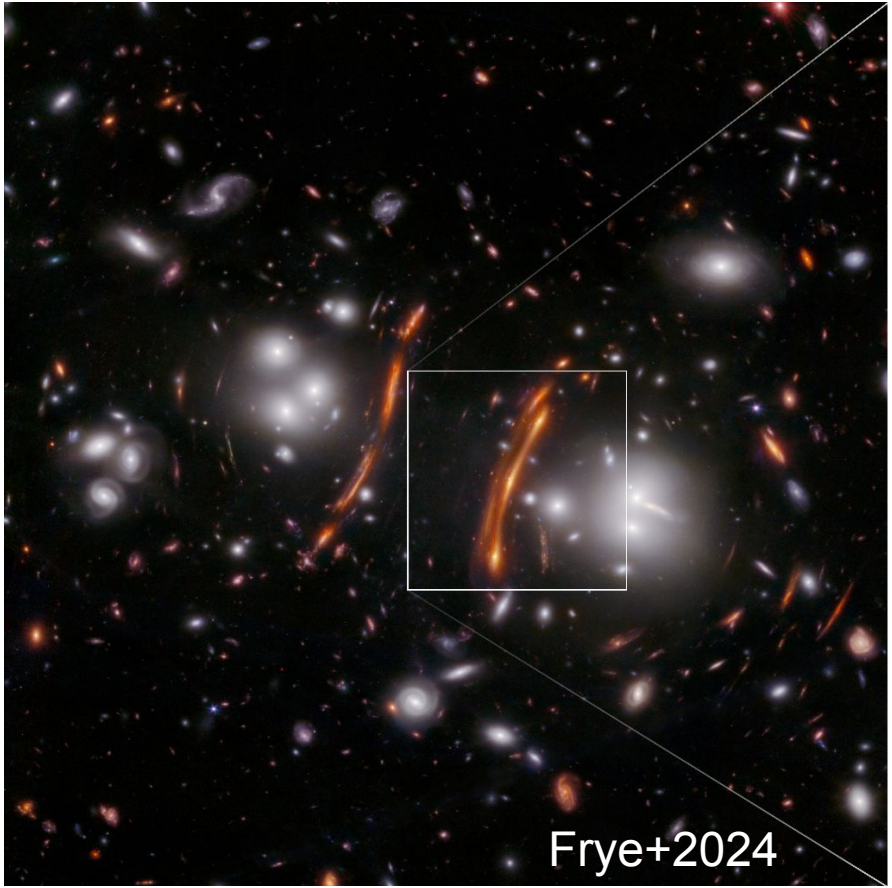


Grayling+2025

Constraining H0

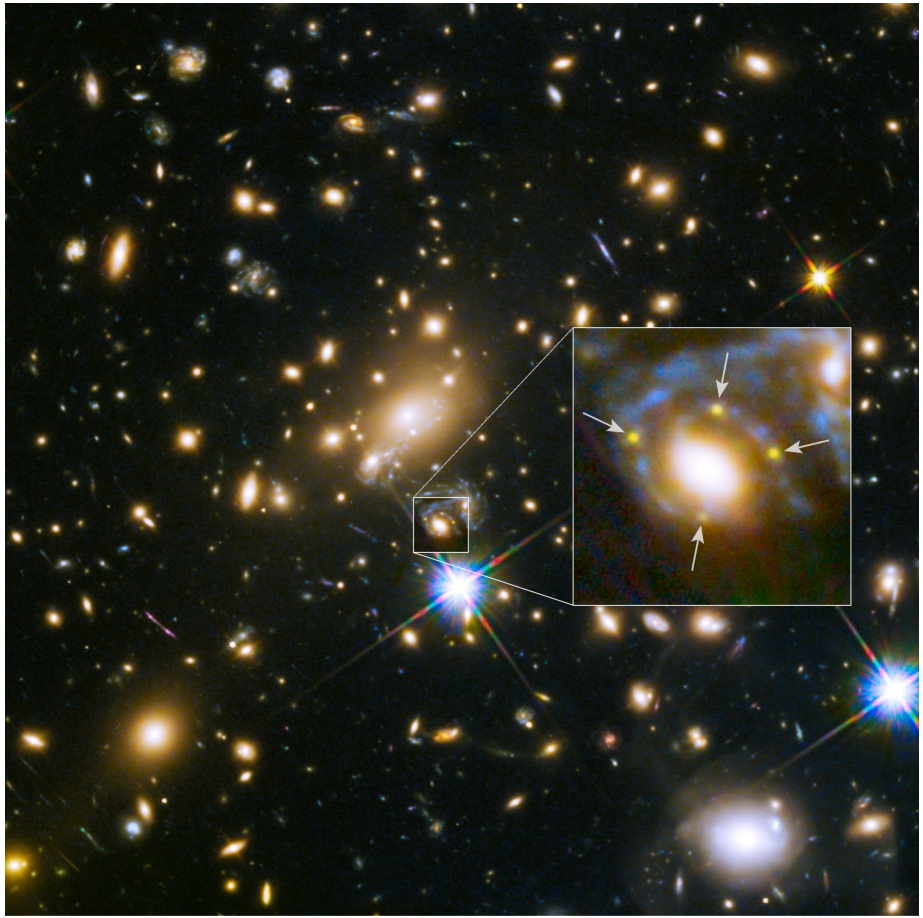


Grayling+2025



The Future of Lensed Supernovae

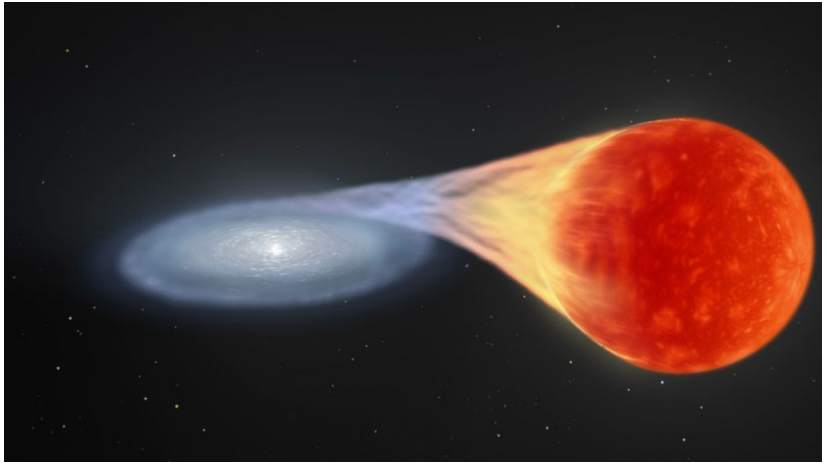
- Strongly-lensed supernovae are a powerful tool to resolve the Hubble tension
- We have only a few currently, but LSST is expected to find hundreds
- Lensed supernovae hold great promise for cosmology, and will hopefully resolve the Hubble tension over the next few years



Conclusions

The Future of Supernova Cosmology

- Supernova cosmology is at an exciting time, but limited by unknown astrophysics
- We can use BayeSN to empirically probe properties of the population of type Ia supernovae
- We plan to infer these properties jointly with cosmological parameters using SBI
- Lensed supernovae are a valuable tool to help resolve the Hubble tension; BayeSN-TD can play a key role



Credit: Adriana Manrique Gutierrez svs.gsfc.nasa.gov